

Software Design Specification

for

The Book Exchange

Version 2.0 approved.

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Revision History

Name	Date	Reason For Changes	Version
First Draft	04/24/24	Initial Draft	1.0
Second Draft	05/01/24	General Revisions	1.5
Final	05/08/24	Finalized revisions	2.0

1. Introduction

1.1 Purpose

A web-based database that helps users in buying and selling books without any hidden or incurring fees of other selling platforms. This system, “The Book Exchange,” shall only keep track of the users, the books they list, the books they purchase, as well as any other relevant user data.

1.2 Scope

The purpose of this document is to provide an overview of the design and development of this project, as well as the recommended procedures and available resources for developers and maintainers of “The Book Exchange.” This document is not an expression of formal policy; it contains documentation for “The Book Exchange” system and generally agreed-upon best practices.

1.3 Intended Audience

This document is for the use of the California State University of Los Angeles (CSULA) Computer Science department. This document will cover the details of the CS 3337 Software Engineering final project for Group 1. The intended audience for this project is the professor and students of the course.

1.4 Product Names/Version Numbers

N/A

1.5 References

- CS 3337 Software Engineering Syllabus – See “Student Learning Outcomes”
- “The Book Exchange” Software Requirements Document

1.6 Definitions, Acronyms and Abbreviations

View Appendix S

2. System Overview

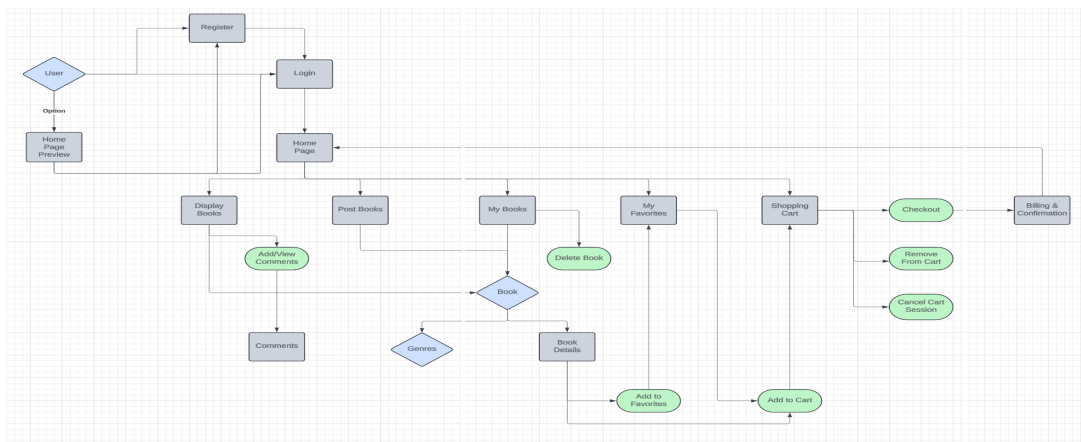
Database management systems (DBMS) are not known to be user-friendly. The proposed user interface program, “The Book Exchange,” shall utilize existing infrastructure to provide a tool for users to access and manage buying and selling books with an online database with more ease and efficiency.

“The Book Exchange” is designed as an enterprise software tool which uses existing infrastructure, primarily the Django framework and SQLite 3. “The Book Exchange” is compliant with all CSULA protocols and policies, as well as industry standards. The tool shall provide various user interfaces which will allow functions such as user registration, posting, purchasing, commenting, and a favorites list.

The benefit of this system over existing marketplaces is it circumvents the seller fees that exist on other platforms, which can lower the overall cost of a book for the buyer and seller. “The Bookshelf” will maintain all the usability of other sites but give more purchasing power back to the people.

This program will allow registered users to:

- View Books
- Write Comments
- Delete Own Comments
- Post Books
- Sell Books
- Search Books by Keywords
- View Shopping Cart
- Purchase from Shopping Cart
- Add to Favorites
- Remove From Favorites
- View Own Book Listings
- Rate Book Listings



3. Design Considerations

3.1 Assumptions and Dependencies

Software and tools used to develop the website include:

- Django (version 3.11)
- Pycharm
- Python
- SQLite3
- HTML5
- CSS
- Bootstrap

3.2 General Assumptions

The website requires that users have access to the internet. Users will also need a browser that is compatible with the website. Furthermore, for users to be granted access to all the website functions, they must have a registered account and be logged in.

3.3 Goals and Guidelines

The goal of our team is to create a community for book enjoyers to gather and express their interest. Our user interface allows users to have a pleasant experience as they either view, sell, or purchase books. There is a support team that is available to users if they have any questions about content being sold, returns/exchanges, and other policies. With the services provided, the team prioritizes user satisfaction.

3.4 Development Methods

The development of “The Book Exchange” was a collaboration of different tools such as Django, Python, HTML5, and Bootstrap. The tool that made this collaboration effective was PyCharm, which made using the tools together seamless. Since Django uses the model-template-view (MTV) pattern, it made aspects such as requesting or uploading data to the database easy and also made it simple for functions to be modified and maintained as the project moved further into development.

4. Architectural Strategies

4.1 Use of Particular Products

- 4.1.1** Web framework - We must use the Django web framework to implement “The Book Exchange.”
- 4.1.2** Database - We must use the SQLite 3 database system alongside Django.
- 4.1.3** Python for basic programming architecture of the modules.

4.2 Reuse of Existing Software Components

- 4.2.1** The software shall allow access to a search bar when looking for specific book selections.
- 4.2.2** A review system for the books shall be a feature for those users who are unsure of making a purchase.
- 4.2.3** The software shall use the Django framework to implement all its components and store data in a database.

4.3 Future plans for software enhancement

Several features have been marked for future development.

- 4.3.1** Search by Genre
- 4.3.2** Transition Comment and Favorite view from table-based to card-based.
- 4.3.3** Order History
- 4.3.4** Contact Us Form
- 4.3.5** Payment Processing System (Stripe, PayPal)
- 4.3.6** External tracking platform (USPA, FedEx, UPS)
- 4.3.7** Newsletters
- 4.3.8** Email Confirmations

4.4 User Interface Paradigms

- 4.4.1** The user interface shall present the home screen upon initial access.
- 4.4.2** The system shall provide a uniform look and feel between all pages. Headings, banners, fonts, and buttons shall follow the same style guide.
- 4.4.3** The navigation bar containing the logo, banner, search bar, login/logout option, Account tab, and About us tab shall be present on every page of the website for easy navigation.
- 4.4.4** The login page shall present the user with two options: login or create an account. Creating an account shall create a new user and credentials in the user database. A logged-in user shall have access only to their own account. Any changes made will update the single user's account in the database. The user shall have access to their own account information (name, address, etc.), purchase history, and review history.
- 4.4.5** The book database shall be updated by posting a book, purchasing a book, commenting, or favoriting a book.
- 4.4.6** The software shall allow administrators to make modifications to the system and the contents of the database through Django's administrator panel.

4.5 Hardware and/or Software interface paradigms

- 4.5.1** Hardware interface paradigm
Device must have a display screen (may or may not be a touch display) and either a keyboard/mouse for user input.
- 4.5.2** Software Interface Paradigm
 - 4.5.2.1** SQLite3

4.6 Error detection and recovery

- 4.6.1** Implement exception handling in the codes.
- 4.6.2** Debugging the software interfaces programs.
- 4.6.3** To resolve issues, a discussion must be held within the group.

4.7 Memory Management Policy

- 4.7.1** The Django Framework shall manage the memory management of the software.

4.8 External databases

- 4.8.1** The software shall use the SQLite 3 database system, which allows for easy migration to other systems.

4.9 Distributed data/ control over a network

N/A

4.10 Generalized approaches to control

4.10.1 Using the admin privileges to manage the software.

4.10.2 The software shall utilize the MTV architecture, which aligns best with Django.

4.11 Concurrences and Synchronization

4.11.1 The Django Framework shall handle concurrency and synchronization.

4.12 Communication mechanisms

4.12.1 The software may allow the user to send a message to the book author.

4.12.2 The software shall allow the user to contact the owner of the site.

4.12.2 The software shall use the HTTP protocols

5. System

Level 0 DFD

Login verification is required to access the database features of The Book Exchange. Without login, public users will only be able to access the homepage, log in, search bar, and about pages.

Level 1 DFD

Public users are allowed access to the Home page, About Us page, and search bar. To be granted access to the website's other pages and functions, the user needs to register an account and log in. If a user does not have an account with The Book Exchange, they will be directed to register a username and password before they log into the website. Django provides the register manager with checks to approve the user's registration. Once a user is registered and logged in, they will have access to the pages that can update the database.

6. Detailed System Design

6.1 User Registration

6.1.1 Definition:

Users establish an account with their credentials to log in to The Book Exchange website.

6.1.2 Responsibilities:

Creating an account shall grant access to all the features of The Book Exchange website.

6.1.3 Constraints:

Username must be 150 characters or fewer, with only letters, digits, and @/./+/_ only.

Users' passwords will follow the constraints: they cannot be like other personal information, they must contain at least 8 characters, they cannot be a commonly used password, and they cannot be entirely numeric.

6.1.4 Composition:

Creating an account requires a username and password, as well as verification of the password.

6.1.5 Uses/Interactions:

Upon user registration, the user will be able to log in with their credentials and access The Book Exchange website.

6.1.6 Resources:

NA

6.1.7 Processing:

Once the user has given a valid username and password, their information shall be stored in the database. The user can use their credentials to log in to The Book Exchange website.

6.1.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.2 Login

6.2.1 Definition:

Users will input the username and password they used during account registration.

6.2.2 Responsibilities:

The login process is the key to accessing The Book Exchange website.

6.2.3 Constraints:

Users are required to have created an account before the login process.

6.2.4 Composition:

All logins are composed of a username and password text field.

6.2.5 Uses/Interactions:

After a user has logged in successfully, they shall be sent to The Book Exchange homepage and have access to all the websites' services.

6.2.6 Resources:

NA

6.2.7 Processing:

During a login attempt, the system shall verify that the username and password match in the database. If the username and password do not match, the user will have to try again.

6.2.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.3 Post Book

6.3.1 Definition:

Users can post books onto The Book Exchange for others to purchase.

6.3.2 Responsibilities:

To take the information about the book the user is posting.

6.3.3 Constraints:

Users must be logged in to their account to access the Post Book page.

6.3.4 Composition:

Posting a book requires the book to have: Name, Website URL, Price, and Picture.

6.3.5 Uses/Interactions:

Posted books shall interact with all the main features of the website, including: Display books, My books, Search Books, Shopping Cart, and Book details.

6.3.6 Resources:

NA

6.3.7 Processing:

The name, website URL, price, and picture submitted by the user shall be taken and stored in the database.

6.3.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.4 Display Books

6.4.1 Definition:

Allows users to view all books stored in The Book Exchange database.

6.4.2 Responsibilities:

To display all the books stored in the database.

6.4.3 Constraints:

Users must be on ‘The Book Exchange’ website.

6.4.4 Composition:

The Display Books page shall show the name, image, price, date posted, the user who posted, and current availability of the book.

6.4.5 Uses/Interactions:

Display books will interact with the book model in the database to display books, the book storing method, and book details.

6.4.6 Resources:

NA

6.4.7 Processing:

Display Books shall fetch the book information to be displayed from the database.

6.4.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.5 Book Detail

6.5.1 Definition:

Allows users to view detailed information about a specific book. You can also add the book to the cart or your favorites.

6.5.2 Responsibilities:

To display detailed information about a book in the database.

6.5.3 Constraints:

Users must be logged in to their account to access the Book Details page.

6.5.4 Composition:

The Book Details page shall show the name, image, price, date posted, and what genre the book belongs to. From this page, users will also be able to add said book to the shopping cart and also be able to add to their favorites.

6.5.5 Uses/Interactions:

Book details shall interact with the search bar, display, shopping cart, favorites, and the book model in the database.

6.5.6 Resources:

NA

6.5.7 Processing:

Book details will fetch book information and user reviews from the database.

6.5.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.6 My Books

6.6.1 Definition:

Allows users to view and delete the books they have posted to The Book Exchange.

6.6.2 Responsibilities:

To display the information about the books a user has posted and allow users to delete their posted books from the database.

6.6.3 Constraints:

Users must be logged in to their account to access the book detail page. Furthermore, users must have posted a book into the database to view content on the “My Books” page.

6.6.4 Composition:

My Books page shall show users the names of the books and dates the books were posted to The Book Exchange. It shall also display their username and allow the user to click a link to delete any of their posted books from The Book Exchange.

6.6.5 Uses/Interactions:

My Books shall interact with the book model in the database in the case of a user wanting to delete a book from their posted books.

6.6.6 Resources:

NA

6.6.7 Processing:

My Books shall fetch a user's book information from the book model in the database and display it to the user. When a user is making a book deletion, it shall be deleted based on the identification number in the database.

6.6.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.7 Search Books

6.7.1 Definition:

It allows users to type keywords into a search bar and search the database for any books that match it.

6.7.2 Responsibilities:

To pull from the database any books found using the keywords entered by users.

6.7.3 Constraints:

Users do not have to be logged in to their account to access the search engine and must type keywords into the search bar to be more specific in the search.

6.7.4 Composition:

The search bar is a text field at the top of every webpage of The Book Exchange that shall accept letters, numbers, and symbols on a keyboard.

6.7.5 Uses/Interactions:

Search books shall interact with a results page when books are found or not found. From the results page, a user can select a book, redirecting them to its book detail page.

6.7.6 Resources:

NA

6.7.7 Processing:

Search books shall take the input submitted by the user and match them with keywords in the Book model database. Searches are not case-sensitive and shall display any books if there are keywords that match the user's search request. The search engine shall look through the book names and the names of the users who posted books and display the results on the result page if any matches are found.

6.7.8 Interface/Exports:

- Python
- Django

- SQLite 3

6.8 Comments

6.8.1 Definition:

Allows users to leave or read a comment on the display book page.

6.8.2 Responsibilities:

To allow a user to read past comments or leave a comment so that other customers can view opinions.

6.8.3 Constraints:

Users must be logged in to their account to access the comment feature.

6.8.4 Composition:

The comment feature involves a text field from which users can type a review and submit it to the database to be saved. Comments displayed on the webpage take the form of a text field and a delete button if you are the user who wrote the comment.

6.8.5 Uses/Interactions:

Comments shall interact directly with the book model in the database to attach reviews to correct books. Comments also interact with the Display Books page since any given book shall be populated with appropriate customer reviews.

6.8.6 Resources:

NA

6.8.7 Processing:

Review books shall take words typed into the text field at the bottom of the book detail page and, once submitted will store them in the database. Book detail shall also populate each book with its corresponding set of reviews stored in the database.

6.8.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.9 Favorites

6.9.1 Definition:

Allows users to add books to their favorites list.

6.9.2 Responsibilities:

To allow a user to add books to their favorite list and allow the user to view all favorites.

6.9.3 Constraints:

Users must be logged in to their account to access the favorites feature.

6.9.4 Composition:

The user must be logged into “The Book Exchange.”

6.9.5 Uses/Interactions:

The Favorites page shall interact with the user details page and the users stored in the database.

6.9.6 Resources:

NA

6.9.7 Processing:

Favorites page shall fetch the names, images, and prices of books stored in the database and populate the page with each book.

6.9.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.10 Shopping Cart

6.10.1 Definition:

Users will be able to view books that they have added to the shopping cart from this page.

6.10.2 Responsibilities:

This will allow users to view the total cost and other information about their books before purchasing them.

6.10.3 Constraints:

Users must be registered and logged in to their account to access the ‘shopping cart’ feature.

6.10.4 Composition:

The Shopping Cart shall include details about all books added to the shopping cart, such as title and price.

6.10.5 Uses/Interactions:

“Shopping Cart” interacts with the “Purchase” page, which allows the user to complete their purchase and mark books as sold. Books can also be added and removed from the shopping cart from the “Book Detail” page.

6.10.6 Resources:

NA

6.10.7 Processing:

Once a book has been added to the shopping cart, the user who added the book will be able to purchase the book from their shopping cart.

6.10.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.11 Checkout

6.11.1 Definition:

Users are purchasing books from the website.

6.11.2 Responsibilities:

Make sure users can purchase books that have been put into the listing and send confirmation emails when books are purchased.

6.11.3 Constraints:

Users must be logged in to purchase books and have books in their shopping cart, the book must be listed as available.

6.11.4 Composition:

Uses Book model to update the database that the book is purchased.

6.11.5 Uses/Interactions:

This page is connected directly to the shopping cart page.

6.11.6 Resources:

NA

6.11.7 Processing:

The system shall access the internet to process the payment and send the confirmation email.

6.11.8 Interface/Exports:

- Python
- Django
- SQLite 3

6.12 About Us Page

6.12.1 Definition:

Any information regarding our website and its services can be found on the “About Us” page.

6.12.2 Responsibilities:

This page was designed to inform our users of our goals and services. Our contact information can be found on this page if the user has any questions or concerns.

6.12.3 Constraints:

Both public and registered users may have access to this page.

6.12.4 Composition:

This page displays detailed services and contact information.

6.12.5 Uses/Interactions:

This page can be accessed from the “About” tab located at the top of the website.

6.12.6 Resources:

NA

6.12.7 Processing:

NA

6.14.8 Interface/Exports:

- Python
- Django

7. Graphical User Interface Design

Homepage

The homepage is the website's landing page and will refresh daily to showcase books with a randomly selected genre. Public users will have a limited view of navigation options. Registered and logged-in users will have access to all the navigation tabs and their shopping carts.

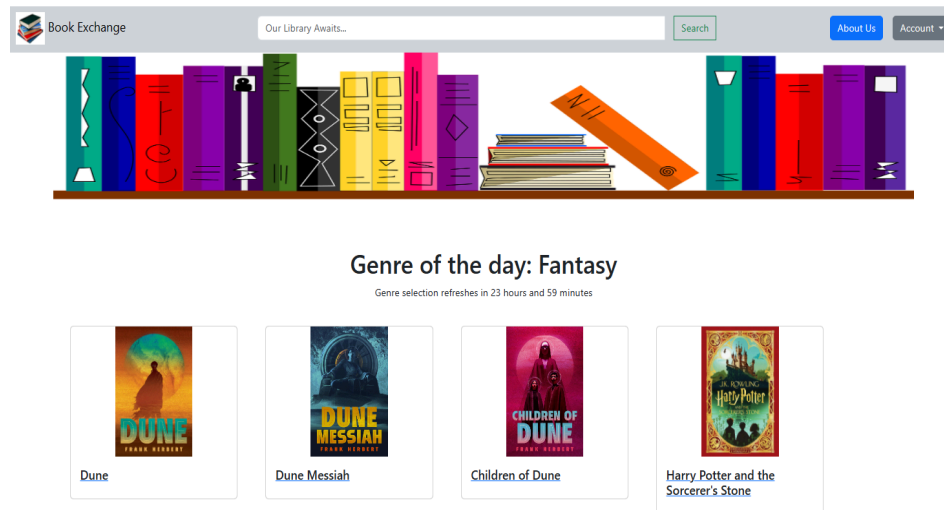
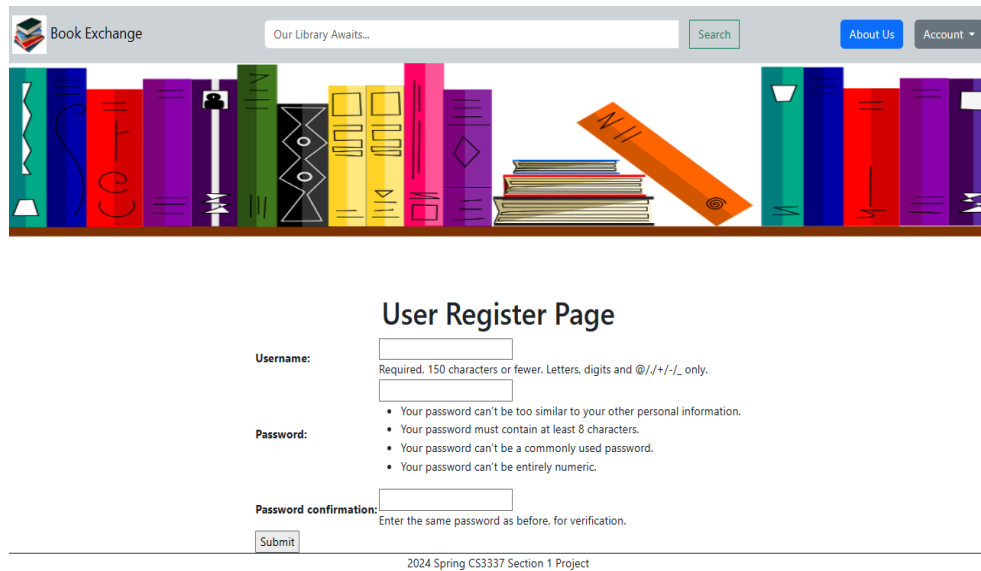


Figure 2 – Home Page

Register User

The user registration page makes use of the Django registration form. Users shall be able to create a new account, log in, and use website functions.



Book Exchange Our Library Awaits... Search About Us Account

User Register Page

Username:
Required. 150 characters or fewer. Letters, digits and @/./+/-/_ only.

Password:

- Your password can't be too similar to your other personal information.
- Your password must contain at least 8 characters.
- Your password can't be a commonly used password.
- Your password can't be entirely numeric.

Password confirmation:
Enter the same password as before, for verification.

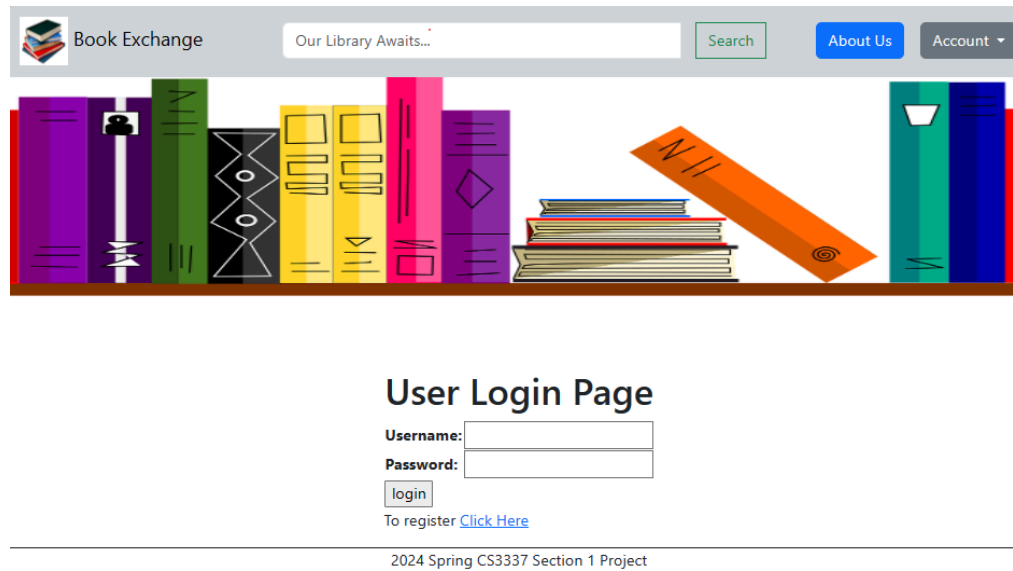
Submit

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Figure 3 – Register User

Login/Logout

The login page allows users to log in and be redirected to the homepage, which now has access to website functions. Unauthorized users will be redirected to the login page. Logout is available in the navigation bar.



The image shows the header of a website called "Book Exchange". The header includes a logo with a stack of books, the text "Book Exchange", a search bar with the placeholder text "Our Library Awaits...", a "Search" button, an "About Us" button, and an "Account" dropdown menu. Below the header is a decorative banner featuring a row of colorful books (purple, green, black, yellow, pink, purple) and a stack of books. Below the banner is the "User Login Page" which contains a "Username:" label, a text input field, a "Password:" label, another text input field, a "login" button, and a link "To register [Click Here](#)". At the bottom of the page, there is a footer with the text "2024 Spring CS3337 Section 1 Project".

Book Exchange

Our Library Awaits...

Search

About Us

Account

User Login Page

Username:

Password:

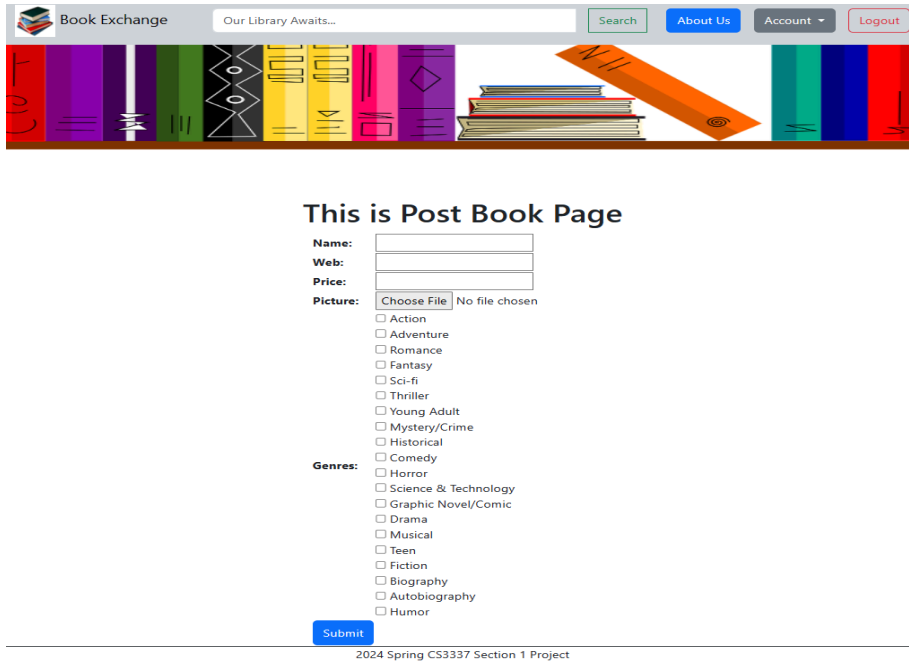
To register [Click Here](#)

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Figure 4 – Login

Post Books

The Post Book page allows users to put books up for sale. Prompts the user for information and once submitted will be visible in the Display Books & Home page.



Book Exchange Our Library Awaits... Search About Us Account Logout

This is Post Book Page

Name:

Web:

Price:

Picture: No file chosen

☐ Action

☐ Adventure

☐ Romance

☐ Fantasy

☐ Sci-fi

☐ Thriller

☐ Young Adult

☐ Mystery/Crime

☐ Historical

☐ Comedy

Genres:

☐ Horror

☐ Science & Technology

☐ Graphic Novel/Comic

☐ Drama

☐ Musical

☐ Teen

☐ Fiction

☐ Biography

☐ Autobiography

☐ Humor

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Figure 5 – Post Books

Display Books

Display Books allows users to view all books that can be purchased. Users can click on links to view more details about the book and leave reviews. Display Page link is under the Account tab.



Display Books

Name	Picture	Genre(s)
Django 2.3		Add Comment View Comments
Django 4.0		Add Comment View Comments

Figure 6 – Display Books

Book Detail

When users click on a book name from the Display Page or from the Search Results page, they are redirected to its Book Details page. The page allows users to see more details about a book. Users can add the book to their shopping cart or favorites list through each respective button.

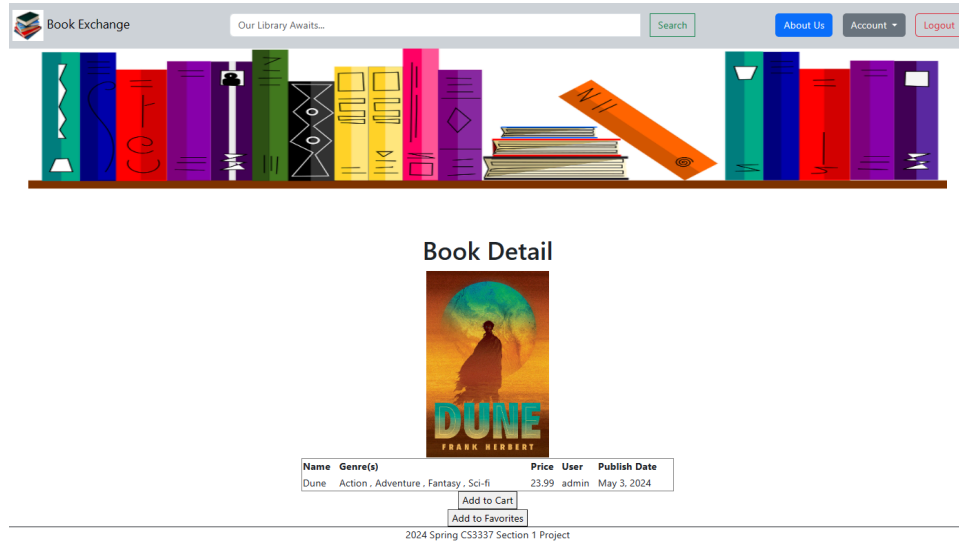
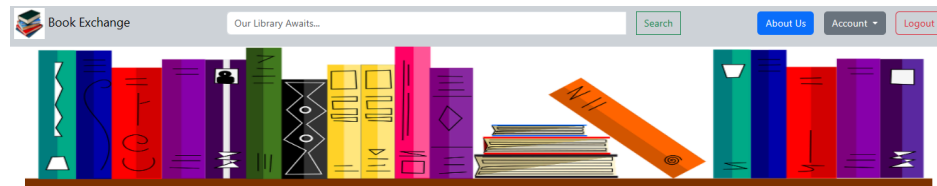


Figure 7 – Write a Review

My Books

The My Books page link is under the Account tab, which will allow users to view and delete the books that they have posted for sale.



My Books

Name	Genre(s)	Picture	DELETE
Django 2.3			Delete
Dune	Action , Adventure , Fantasy , Sci-fi		Delete
Dune Messiah	Action , Adventure , Fantasy , Sci-fi		Delete
Children of Dune	Action , Adventure , Fantasy , Sci-fi		Delete
Harry Potter and the Sorcerer's Stone	Adventure , Fantasy , Young Adult		Delete

Figure 8 – My Books

Search Books

The search bar feature allows users to search all books with keywords and redirect them to a results page.

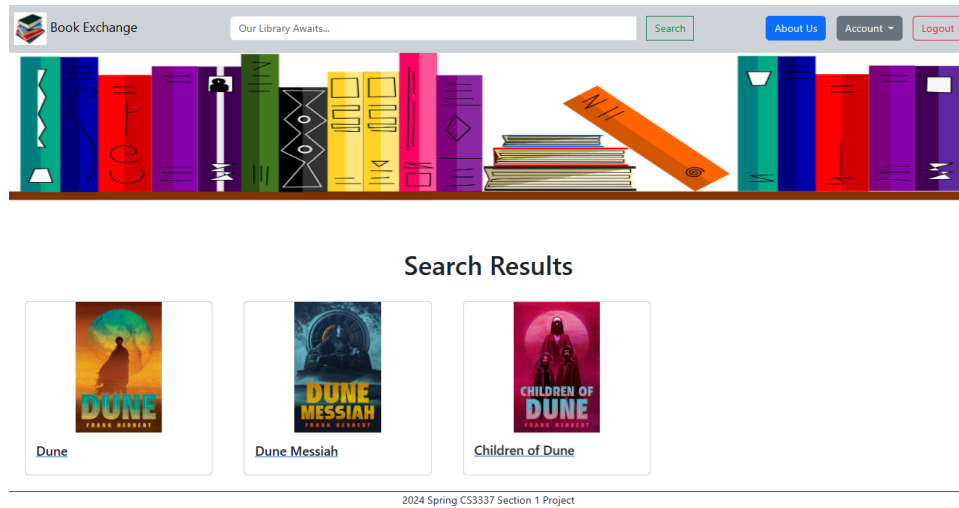


Figure 9 – Search Books

Favorites

Favorites allow users to see a favorites list of books based on what the user has added to their favorites.

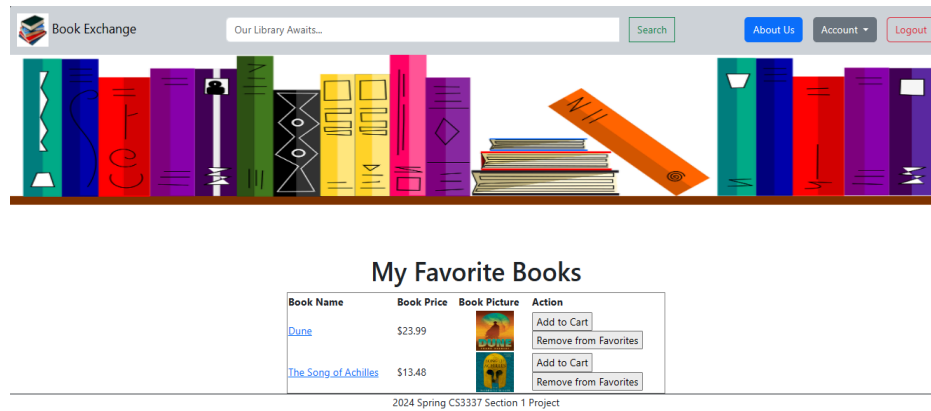


Figure 10 – Sort Books

Comments

All registered users can leave comments. Comments can be added or viewed by their respective buttons on the Display Books page. Comments only contain the description and only be deleted by the user who wrote it.



Comments

Good Read	Delete
Highly Recommend!	Delete
Can't wait to read the next book.	Delete

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Figure 11 – Leave Book Review

Shopping Cart

Users can add books to the shopping cart through the book details page. Multiple users can store the same book in their shopping cart.

The screenshot shows the 'Book Exchange' website's shopping cart. The header includes the site logo, a search bar with the placeholder 'Our Library Awaits...', and navigation links for 'Search', 'About Us', 'Account', and 'Logout'. Below the header is a decorative banner featuring a row of colorful books and a stack of books. The main content area is divided into two sections. On the left, under the heading '(2) items Items in your cart', there is a list of items: 'Dune Messiah' priced at \$23.99 and 'The Song of Achilles' priced at \$13.48. Each item has a quantity of 1 and links to 'Add gift package' and 'Remove item'. On the right, under the heading 'Cart Summary', the total is displayed as '\$37.47'. Below this, a note states: '*For United States, France and Germany applicable sales tax will be applied'. At the bottom of the summary section are buttons for 'Checkout', 'Cancel', and 'Support', along with a phone number '+43 100 783 001'.

Items in your cart	
Dune Messiah Add gift package Remove item	\$23.99 1
The Song of Achilles Add gift package Remove item	\$13.48 1

Cart Summary
Total
\$37.47
*For United States, France and Germany applicable sales tax will be applied
[Checkout](#) [Cancel](#)
[Support](#)
+43 100 783 001

Figure 13 – Shopping Cart

Checkout

If the book is available, then it can be purchased by users from their shopping cart. To make a purchase users must fill out their address and payment method information.

The screenshot shows the 'Book Exchange' website's checkout page. The header is identical to the shopping cart page. Below the header is the same decorative banner. The main content area is divided into two sections. On the left, under the heading 'Checkout', there are form fields for 'Name' (filled with 'John Rivera'), 'Email' (filled with 'JohnRivera@gmail.com'), and 'Address' (filled with '123 Highland Ave'). Below these is a 'Payment Method' section with a 'Credit Card' option. At the bottom of this section is a blue 'Place Order' button. On the right, under the heading 'Order Summary', there is a table listing the items in the cart: 'Dune Messiah x 1 \$23.99' and 'The Song of Achilles x 1 \$13.48', with a 'Total \$37.47' at the bottom.

Checkout	
Name	John Rivera
Email	JohnRivera@gmail.com
Address	123 Highland Ave
Payment Method	Credit Card
Place Order	

Order Summary

Dune Messiah x 1 \$23.99
The Song of Achilles x 1 \$13.48
Total \$37.47

Figure 14 – Checkout

About Page

The About Us page explains the purpose and goals of the website.

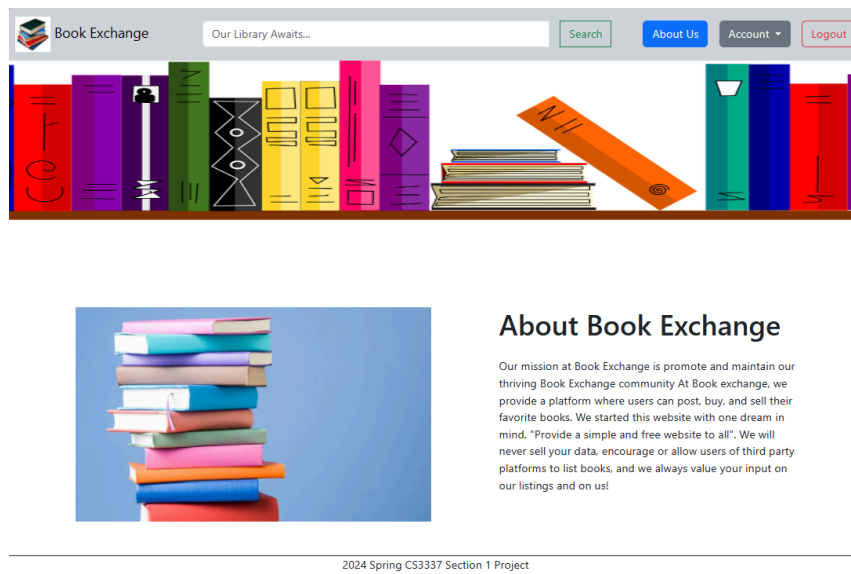


Figure 15 – About

S. Glossary

DFD: Data Flow Diagram

HTTPS: Hypertext Transfer Protocol Secure; internet protocol that is used for secure communication over computer networks.

Django: Python web development framework.

SQLite 3: The database system used by Django.

MTV Architecture: Model Template View Architecture that is commonly used in the Django Framework. Models correspond to data stored in the database, Views correspond to the processing of data, and Templates correspond to the information displayed to the user.

CSS: Cascading Style Sheets

HTML: Hyper Text Markup Language

GUI: Graphical User Interface